

PREM TIWARI

Full-Stack Developer & AI | BTech CSE (AI)

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SUMMARY

BTech CSE (AI) student with a Diploma in Information Technology and a UI/UX web design internship, building production-style full-stack applications and AI-powered learning systems. Comfortable across React/TypeScript front-ends, FastAPI and Node.js backends, Gemini AI integrations, real-time Socket.io layers, and Docker-based delivery. Seeking internships in full-stack development or AI/ML.

TECHNICAL SKILLS

Languages: Python (Advanced), JavaScript, TypeScript

Frontend: React, HTML, CSS, Vite, Responsive Design, UI/UX

Backend: Node.js, Express, FastAPI, REST APIs, Socket.io / WebSockets

Databases: Prisma ORM, PostgreSQL, SQLite, Database Design

AI / ML: Google Gemini AI, TensorFlow/Keras, CNNs, Transfer Learning, OpenCV, Grad-CAM

Tools: Docker & Compose, Git & GitHub, Streamlit, JWT, Third-party APIs

Interests: Cyber Security, System Architecture

EXPERIENCE

Web Design (UI/UX) Intern — Nirvana Tech, Vadodara

Apr 2026 - May 2026 · 5 weeks

- Designed and built responsive web page layouts, converting design mockups into working HTML/CSS interfaces that held up across screen sizes.
- Applied UI/UX fundamentals — visual hierarchy, typography, spacing and consistent components — to improve clarity and usability of assigned pages.

PROJECTS

AI Study Helper — AI-Powered Learning & Interactive Simulation Platform

React 18, FastAPI, Python, Google Gemini AI, SQLite, JWT, Tailwind CSS

- Built a full-stack AI tutor with note/PDF summarization, concept simplification, MCQ quiz generation, personalized study plans and Socratic tutoring.
- Added a YouTube lecture analyzer plus interactive system-failure simulations and a visual logic builder to support multi-modal, hands-on learning.
- Hardened the application through a 12-point security audit covering JWT auth, sliding-window rate limiting, prompt-injection defenses and magic-byte PDF validation.

RentApp — Full-Stack PropTech & Rental Management Platform

React 19, TypeScript, Node.js, Express, Prisma ORM, Socket.io, Docker

- Architected a multi-role platform (tenant, owner, admin) with route guards enforced on both client and server, plus passwordless phone-OTP login.
- Built a real-time layer with Socket.io for sub-second chat, booking updates and emergency broadcasts, using room-based scoping per property and user.
- Delivered end-to-end lease onboarding: canvas e-signatures, multi-app UPI checkout with instant receipts, QR visitor passes and maintenance ticketing.
- Modelled related entities in Prisma with strict foreign-key constraints and containerised the multi-tier stack using Docker Compose multi-stage builds.

Skin Cancer Classification System — AI-Powered Computer-Aided Diagnosis

Python, TensorFlow/Keras, OpenCV, Transfer Learning, Grad-CAM, Streamlit

- Trained a CNN via transfer learning on HAM10000 (10,015 dermoscopic images) to classify lesions as benign or malignant — 92.4% accuracy, 0.94 ROC-AUC.
- Handled class imbalance and applied preprocessing plus augmentation (rotation, flip, zoom); evaluated with precision, recall, F1 and ROC-AUC.
- Added Grad-CAM explainability and shipped a Streamlit app that returns a prediction with a confidence score for any uploaded lesion image.

ScrapBajar — Scrap Collection Platform

Python, Web Development

- Web platform for requesting scrap pickup by location, with pricing driven by current market rates — built around sustainability and digital convenience.

EDUCATION

BTech, Computer Science Engineering (Artificial Intelligence)

Currently pursuing · Expected 2027

Diploma in Information Technology

Parul University, Gujarat, India · 2024